

MASTER REPORT

Summit Care Windows Server 2022 Enterprise Domain Configuration

This document explains how Summit Care Medical Clinic implemented a practical Windows Server 2022 enterprise domain environment. The project covers Active Directory Domain Services, DNS, DHCP, NPS/RADIUS, AD Certificate Services, WDS, Group Policy, Windows Time, backup domain controllers, Synology NAS storage, Proxmox, Hyper-V, VDI, Task Scheduler automation, Event Viewer, SNMP monitoring, Microsoft Entra hybrid identity, and Microsoft 365 SSO.

WINDOWS SERVER 2022**AD DS / DNS / DHCP****NPS & AD CS****PROXMOX / HYPER-V****M365 SSO**

Project Goal

Build a real enterprise domain control plane for identity, network services, secure access, deployment, backup, monitoring, and remote work in a healthcare clinic environment.

Implementation Focus

Move beyond theory by documenting how each Windows Server feature was implemented, validated, and tied to realistic Summit Care workflows.

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OVERVIEW

Purpose And Platform Choice

Summit Care selected Windows Server 2022 because it is stable, widely supported, and well suited for AD DS, DNS, DHCP, NPS, AD CS, WDS, Group Policy, Windows Time, and Hyper-V. Windows Server 2025 may be useful later, but 2022 reduces compatibility risk and avoids early-release bugs during a healthcare infrastructure build.

- Centralize users, computers, groups, service accounts, and policies in Active Directory.
- Provide reliable DNS and DHCP so clients can find domain services and internal tools.
- Use GPOs to secure workstations, deploy certificates, map resources, and enforce baselines.
- Secure office network access with NPS/RADIUS, 802.1X, WPA3-Enterprise, and AD CS.
- Support remote work and testing with VDI, VPN, RDP, Hyper-V, and Proxmox virtual machines.
- Build resilience through a backup domain controller, replication, Synology NAS backups, and scheduled automation.

FOUNDATION

Active Directory, Domain Controllers, DNS

And Windows Time

System	Example Name	Role
Primary domain controller	SC-DC01	Static IP, AD DS, DNS, DHCP, GPO management, and primary time hierarchy role.
Backup domain controller	SC-DC02	AD DS, DNS, global catalog, replication partner, and failover authentication.
NPS/RADIUS server	SC-NPS01	802.1X, WPA3-Enterprise, user/device network authentication, and RADIUS logs.
Certificate authority	SC-CA01	Enterprise CA for computer certificates, NPS certificates, and auto-enrollment.
Deployment server	SC-WDS01	PXE boot, Windows images, and standardized workstation deployment.

Domain Controller Implementation

1. Assign static IP addresses to SC-DC01 and SC-DC02 and exclude all server IPs from DHCP scopes.
2. Install AD DS and DNS on SC-DC01, create the Summit Care domain, and organize OUs for users, workstations, servers, VDI, service accounts, and groups.
3. Join SC-DC02 to the domain, promote it as a secondary domain controller, install DNS, and validate SYSVOL and NETLOGON.
4. Run dcdiag and repadmin /replsummary to confirm domain health and replication.
5. Configure DHCP option 006 so clients receive both domain controllers as DNS servers.
6. Verify Windows Time with w32tm because Kerberos, certificates, logs, and NPS authentication depend on accurate time.

DNS Records For Internal Services

Record	Purpose	Benefit
dc01 / dc02	Domain controller and DNS discovery.	Clients locate authentication and Group Policy services reliably.
sql.summitcare.local	Database or clinical tool backend.	Applications use a stable name instead of a changing IP.
intranet.summitcare.local	Internal web server or documentation portal.	Staff and IT reach internal tools by friendly DNS name.
nas.summitcare.local	Synology backup and shared storage.	Scripts and users can target protected storage consistently.

SERVICES

DHCP, WDS, Synology NAS And Backup Strategy

DHCP provides office clients, VDI systems, printers, phones, and test machines with managed addressing. Critical systems use static addresses or reservations so domain services, NAS storage, RADIUS, WDS, and database tools remain predictable.

Service	Configuration	Why It Matters
DHCP scope	Client pool, exclusions, reservations, option 003 router, option 006 DNS, option 015 suffix.	Fast onboarding and fewer manual IP mistakes.
DHCP failover	Windows DHCP failover or documented emergency edge DHCP with non-overlapping pools.	Clients can still receive leases during service failure or maintenance.
WDS	PXE boot, boot images, install images, and pilot deployment to a virtual desktop first.	Standardizes workstation rebuilds and reduces manual install time.
Synology NAS	Domain joined, permissioned with AD groups, used for backups and retained exports.	Keeps recovery data separate from the production server disks.

Backup Targets

- System state backups for domain controllers.
- GPMC exports for Group Policy Objects.
- DHCP configuration and lease exports.
- AD CS database, private key, certificate templates, and CA configuration backups.
- PowerShell scripts, Task Scheduler XML exports, and operational documentation.
- Hypervisor VM backups or exports for domain services, VDI, and test machines.

SECURITY

Group Policy, NPS/RADIUS, Certificates And

Entra SSO

Control	Summit Care Implementation	Security Value
Group Policy	Password rules, lockout policy, workstation firewall, mapped drives, RDP control, certificate auto-enrollment, and Wi-Fi profiles.	Repeatable security and configuration across domain devices.
NPS/RADIUS	RADIUS policies for wireless access points, switches, user groups, computer groups, and approved authentication methods.	Network access is tied to real identity instead of a shared password.
802.1X / WPA3-Enterprise	Domain users and computers authenticate to the internal office network through NPS.	Lost or disabled accounts can be blocked without changing an office-wide Wi-Fi key.
AD Certificate Services	Enterprise CA, computer certificate templates, NPS server certificate, auto-enrollment GPO, CRL/AIA planning.	Trusted domain computers can receive certificate-based access automatically.
Microsoft Entra SSO	Verified public domain, UPN suffix setup in Active Directory Domains and Trusts, Entra sync, MFA, and Microsoft 365 app sign-in.	Users sign into M365 apps with a clean business identity such as user@summitcare.org.

A real GPO scenario: domain computers receive certificates through auto-enrollment, receive the secure office Wi-Fi profile through Group Policy, validate the NPS server

certificate, and connect through WPA3-Enterprise / 802.1X without using a shared Wi-Fi password.

OPERATIONS

Proxmox, Hyper-V, VDI, Automation And Monitoring

A type 1 hypervisor lets Summit Care use one physical server for multiple roles: domain controllers, NPS, CA, WDS, virtual office desktops, remote work VDI, and sandbox testing. Proxmox hosts the core virtual machines, while Windows Hyper-V and nested virtualization can be used for Microsoft lab testing where supported.

Operational Area	Implementation	Benefit
VDI / remote work	Remote staff VPN into the network, then RDP into assigned virtual office desktops.	Internal applications stay inside the controlled clinic environment.
Sandbox testing	Virtual desktop clients receive GPOs, certificates, DNS, DHCP, and app access before production rollout.	Changes can be validated before touching physical clinic workstations.
Task Scheduler	Scheduled scripts export DHCP, backup GPOs, run dcdiag/repadmin, copy outputs to NAS, and export logs.	Routine maintenance becomes repeatable and documented.
Event Viewer	Review Directory Service, DNS, DHCP, NPS, certificate enrollment, security, and system logs.	Faster troubleshooting and better audit evidence.
SNMP monitoring	Monitor switches, APs, NAS, UPS, hypervisor host, and other infrastructure devices.	Visibility into uptime, efficiency, and device health.

SUMMARY

Final Benefits

The completed Summit Care Windows Server 2022 implementation turns the clinic server environment into a full enterprise platform. Active Directory centralizes identity. DNS makes internal tools discoverable. DHCP gives clients reliable addressing. A backup domain controller protects logon and name resolution. Group Policy secures workstations at scale. NPS/RADIUS and AD CS enable enterprise network authentication and certificates. WDS standardizes desktop deployment. Synology NAS supports the backup strategy. Proxmox, Hyper-V, and VDI make the hardware flexible for testing and remote work. Task Scheduler, Event Viewer, SNMP, and health checks improve daily operations. Microsoft Entra hybrid identity and UPN alignment connect the local domain to Microsoft 365 SSO. The result is a healthcare network that is easier to manage, easier to recover, and far more secure than unmanaged workgroup computers or shared-password network access.